Questionnaire: Query-Expansion + Reranking RAG over Tiffinly Product Documents

Answer every question in your own document, labelled with its ID (A1, B2 ...). Total: 100 marks, scaled to the 35 questionnaire marks in the brief's grading table.

How answers are marked. A correct answer that is not backed by something you observed in your own runs gets at most half marks. "Evidence" means a number copied from your terminal, a chunk's source file and page, a rank, a score or a screenshot. Concept answers (Section A) should be 2 - 4 sentences in plain language; no copying from the brief.

Section A - Concepts (answer before you write code)

ID	Question	Marks	Evidence to attach
A1	In one or two sentences, what problem does RAG solve that asking GPT-4o directly does not?	2	-
A2	Why do we split PDFs into chunks instead of embedding each whole page or whole document?	2	-
A3	What does CHUNK_OVERLAP protect against? Give an example of a fact in the Tiffinly PRD that could be damaged without it.	3	PRD page + fact
A4	A FAISS search and a cross-encoder both produce a relevance judgement. Explain how each one gets its judgement, and why the cross-encoder is not used to search the whole index.	3	-
A5	Query expansion is said to improve *recall* and reranking to improve *precision*. Define both words for this pipeline, using "right chunk" and "top 5" in your definitions.	3	-
A6	Why must ingest.py and retriever.py use the same embedding model? What would you see if they did not?	2	-
A7	Why does the answer prompt force the model to reply with an exact refusal sentence when the context has no answer?	2	-

Section B - Ingestion (Step 1, ingest.py)

ID	Question	Marks	Evidence to attach
B1	Report the number of PDFs, pages, chunks and vectors printed by your ingest.py. Do chunks and vectors match? Why must they?	2	Terminal output
B2	Run ingest.py twice. Are the counts identical? Which part of the pipeline would make results differ if you ran the whole system twice, and which part never changes between runs?	2	Two outputs
B3	Paste your sample chunk. It begins with the page footer and "Page 1", and table cells appear one per line. How could the repeated footer text hurt retrieval? How would you remove it, and in which function?	4	Sample chunk
B4	Open the faiss_index folder. What is stored in index.faiss and what in index.pkl? Why is allow_dangerous_deserialization=True acceptable here but dangerous for a file downloaded from the internet?	3	-

Section C - Baseline retrieval (Step 2, retriever.py)

Run all nine questions (Q1 - Q9) from Section 10 of the brief, both wordings, through retriever.py only. Fill in the grid below in notes.md; you will extend it in Sections D and E.

ID	Question	Marks	Evidence to attach
C1	Copy your grid. For each question and wording: is the chunk holding the correct answer in the 6 results (yes / no), and at what rank?	4	Completed grid
C2	Which everyday wordings failed while the document wording worked? For one failure, name the words in the question and the words in the document that did not match.	3	Chunk text
C3	Q5 "Did the new pause feature stop people leaving?" and Q2 "How many subscribers quit each month?" both concern churn. Did the baseline return chunks from the PRD, the roadmap or the readout for each? Which document actually holds each answer?	3	Sources + pages

Question	Wording	Retriever: found? rank	Reranker: found? rank	Full pipeline: found? rank	Answer correct?
Q1	Doc / Everyday				
...	...				
Q9	Everyday	(should be: nothing relevant)			Refusal sentence?

Section D - Reranking (Step 4, reranker.py)

ID	Question	Marks	Evidence to attach
D1	Run Q2 (everyday wording). Paste the BEFORE / AFTER block. Which chunk moved up the most? Give its old rank, new rank and score.	3	Screenshot
D2	Was the correct chunk already in the 6 results before reranking, just lower down? What does this tell you about what reranking can and cannot fix?	3	BEFORE / AFTER
D3	Cross-encoder scores can be negative. Is a score of -2.5 "bad"? What actually matters: the value or the order? Support your answer with two scores from your run.	2	Scores
D4	Set TOP_N = 2, then TOP_N = 8 (with TOP_K still 6). For Q8 (needs two documents), what happened to the answer in each case, and why?	3	Both answers

Section E - Query expansion (Step 5, query_expansion.py)

ID	Question	Marks	Evidence to attach
E1	For Q4 everyday wording, list the 5 phrasings the model generated. Which ones used words that really appear in the roadmap (e.g. "North Star")? Which were useless?	3	Terminal output
E2	Report total searches, total chunks fetched and unique chunks after dedupe for Q4. Explain the gap between "fetched" and "unique" in one sentence.	2	Numbers
E3	Find one question where the full pipeline retrieved the correct chunk and plain retrieval (Section C) did not. Name the chunk (file + page) and the phrasing that found it. This is the main payoff of expansion; show it clearly.	4	Both outputs
E4	Run the same question three times. Do the phrasings and the unique-chunk count change? Which setting causes this, and would you lower it in production? Why or why not?	3	3 runs
E5	The dedupe key is chunk.page_content. Describe a situation where two chunks are the same fact but are not caught as duplicates, and one where this key is exactly right.	2	-
E6	Why does rerank() receive the original question and not one of the expanded phrasings? What could go wrong otherwise?	2	-

Section F - Tuning experiments (change one setting at a time)

Re-run ingest.py after every chunk-setting change. Use Q3, Q5 and Q8 (both wordings) as your test set for this section.

ID	Question	Marks	Evidence to attach
F1	Chunk size / overlap of 300 / 50, 700 / 250 and 1200 / 200: report the chunk count for each and which setting answered Q3 (a single requirement row) and Q8 (needs two documents) best. Explain why size helps one and hurts the other.	4	Table of runs
F2	TOP_K = 3 and TOP_K = 10 with the full pipeline: report unique chunks after dedupe and whether answers changed.	3	Numbers
F3	NUM_EXPANSIONS = 0, 3 and 5. With 0, what does the full pipeline become? At which value did extra phrasings stop adding new unique chunks for your test set?	3	Numbers
F4	From F1 - F3, state the settings you would ship and give one sentence of evidence for each.	2	-

Section G - Answer quality and failure analysis

ID	Question	Marks	Evidence to attach
G1	Q9 (share price): paste the answer. Did it return the exact refusal sentence? Now ask "What was Tiffinly's revenue in FY26?" - does the model invent a number from ARPU and subscribers? Paste the answer and judge it.	3	Answers
G2	Ask "What is the NPS?" with no time period. The roadmap says 41 (Q4 FY26) and the readout says 44 (Q1 FY27). What did your system answer? Is that acceptable? Suggest one change (prompt or retrieval) that would make it handle this better.	3	Answer + chunks

ID	Question	Marks	Evidence to attach
G3	Q8 needs the roadmap (KR1.1 target 7.0%) and the readout (7.3% result). Did the final top 5 contain both? If not, which stage lost the second document?	3	Top 5 list
G4	Q7 (Hyderabad cut-off): the PRD says 9:00 PM is the cut-off and raises OQ-2; the readout sets 8:00 PM for Hyderabad. Which answer did your system give, and which chunk caused it?	2	Answer + chunks
G5	Pick one wrong or weak answer from any run. Trace it: was the cause in chunking, search, expansion, reranking or the prompt? How do you know?	3	Full trace

Section H - Design reflection

ID	Question	Marks	Evidence to attach
H1	Count the model calls for one question in the full pipeline (embedding, expansion, reranker, answer). Which step adds the most waiting time? Measure it with time.perf_counter() if you can.	3	Timings
H2	Give one type of question where query expansion is not worth its cost, and one where it is essential. Use Tiffinly examples.	2	-
H3	If Tiffinly had 5,000 PDFs instead of 3, which two settings or components would you change first, and why?	2	-
H4	Product managers will use this daily. Name two things you would add before handing it to them (for example a check, a log or a user-facing feature) and what problem each one prevents.	2	-

Marks by section

Section	A	B	C	D	E	F	G	H	Total
Marks	17	11	10	11	16	12	14	9	100

Self-check before you submit

Check	Done?
Every answer is labelled with its question ID and backed by evidence from your own runs.	
The C1 grid covers all 9 questions, both wordings, for all three pipelines (retriever, reranker, full).	
Every page number you quote is the real PDF page (metadata['page'] + 1), checked against the PDF.	
Screenshots folder holds: ingest output, one BEFORE / AFTER rerank block, one full-pipeline run, one refusal (Q9).	
Settings were restored to the defaults in the brief after the Section F experiments, and ingest.py re-run.	
Your submission contains no .env file, no API key and no faiss_index/ folder.	
A classmate can clone your repo, add their own .env, and run all five steps without editing any code.	